

AlphaSense — Professor Review Q&A;

NSE/BSE Next-Hour Volatility-Regime Research | Viva Preparation Guide

Key numbers to memorize: 5 assets (RELIANCE, TCS, HDFCBANK, INFY, ICICIBANK) | 83,259 15m rows | Direction FAIL (+0.1pp) | Volatility PASS (130/130 folds, +13.9pp median) | Locked holdout from 2026-08-10 | 256 tests passed | Frozen 9-feature 1h model.

PART A: Project Choice & Basics

Q: 1. Why did you choose this project?

A: Stock prediction is crowded, but most student projects do next-day prediction with look-ahead leakage on US data. I wanted rigorous 1-hour intraday research specifically for NSE/BSE with strict temporal correctness. The interesting part is we proved a **negative** result honestly (direction fails) and then found a **positive** one (volatility passes).

1-liner: To test if news + market data can predict next-hour NSE/BSE moves with research-grade rigor.

Q: 2. What is AlphaSense in one minute?

A: AlphaSense predicts **next-hour volatility-regime** — will the next hour be high-range or low-range — for 5 NSE stocks: RELIANCE, TCS, HDFCBANK, INFY, ICICIBANK, using 15-min Upstox candles. Phase 0–3 proved **direction prediction fails**. Phase 4 proved **volatility prediction passes: 130/130 walk-forward folds won, median +13.9pp over baseline**.

Q: 3. Why NSE/BSE only? Why not US market?

A: 1. Indian retail problem is underserved.
2. NSE has unique constraints: 09:15–15:30 IST session, special sessions, 2024–2026 holidays, IST/UTC confusion — US tutorials ignore this.
3. We built **NSEBSETradingCalendar + IST normalization + session validation** for this.

Q: 4. Why 1-hour horizon? Why not next-day or 5-minutes?

A: 1-hour is tradable for intraday risk management but not pure noise like 1-min. We tested 30m / 1h / 2h / 4h in Phase 4.3: 30m $\approx$ 1h, edge decays monotonically, 4h flips negative for RELIANCE/ICICIBANK. So **1h is frozen** as the sweet spot.

PART B: Difference from Existing Products — MOST IMPORTANT

Q: 5. What is different from Moneycontrol, Bloomberg, Zerodha Streak, TradingView, Tickertape, Sensibull?

A: **1. They are descriptive, we are predictive:** Moneycontrol/TradingView show charts, news, screeners. We output a probability that the next hour will be high-volatility.

2. They ignore leakage, we enforce it: screeners mix future news with past prices. We enforce **published_at <= prediction_time** for every feature.

3. They are black-box tips, we are evaluated: Streak/Sensibull give backtests with no locked holdout. We use expanding walk-forward (130 folds) + locked holdout from 2026-08-10 scored **once, never tuned against**.

4. We pivoted from direction to volatility: everyone claims to predict UP/DOWN. We proved on 83,259 rows that direction has no edge (+0.1pp) and shifted to range-regime (+13.9pp), useful for option buyers and stop-loss sizing.

Q: 6. There is already a US company called AlphaSense. Is this copied?

A: No. The US AlphaSense is a document-search engine for earnings calls/filings using LLMs. Ours is NSE/BSE intraday signal research using market microstructure + FinBERT sentiment + walk-forward evaluation. Same name coincidence, completely different problem, data, and methods. Ours is not a search tool, it is a prediction + evaluation pipeline.

Q: 7. Why will someone use yours instead of Bloomberg?

A: Bloomberg costs ~\$25k/year and is not built for Indian intraday retail with IST calendar handling. Ours is open, reproducible (256 tests), uses free/low-cost feeds (Upstox, Marketaux, GDELT), and gives calibrated probabilities (Brier 0.2025 vs 0.25 baseline) served via local dashboard + shadow mode.

Q: 8. What is your novel contribution?

- A:
1. NSE/BSE temporal + calendar contract for ML.
 2. Honest negative result for direction on 5 assets with pre-registered PASS criteria (≥ 8 folds, $> +2\%$ median, $\geq 60\%$ win).
 3. Frozen 9-feature volatility model: 6 normalized market features + 1h trailing range/return stats.
 4. News-on-top FAIL proof: news adds nothing even on a working volatility baseline.

PART C: Data & Features

Q: 9. What data do you use?

A: Market: 15-min OHLCV from Upstox, **83,259 rows** combined for 5 assets, 2024-01-01 to 2026-09-23. News: 1,500 articles (RELIANCE 1218, TCS 258, others < 15) from Marketaux + GDELT. Social: Reddit sample + Google Trends daily. All mapped to canonical **MarketBar(asset, exchange, timestamp, OHLCV)**.

Q: 10. Why FinBERT? Why not VADER / TextBlob / ChatGPT?

A: VADER is for tweets, not finance — 'bull, bearish, downgrade, rally' need financial context. FinBERT is BERT fine-tuned on financial phrases and gives positive/neutral/negative probabilities. LLM APIs are non-deterministic, costly, and leak future knowledge. FinBERT runs locally and is reproducible.

Q: 11. How do you prevent future information leakage?

A: Golden rule: feature at time t uses only info $\leq t$. Example: 14:00 article usable for 14:15 prediction, 14:30 article not. Future close used **only** for target, never as feature. Scaling/fitting on train only. Walk-forward trains only on past days.

Q: 12. What features does the final frozen model use?

A: 9 features only: **return_15m, return_30m, return_1h, high_low_range, close_open_return, volume_change + trailing range mean/max + return std over past 1h**. No 4h window — ablation showed 4h redundant. No news — news failed.

PART D: Modeling & Evaluation

Q: 13. Why Logistic Regression? Why not LSTM / Transformer?

A: Baseline-first discipline. If simple logistic cannot beat majority-class on 130 folds, a complex LSTM will just overfit. We also tried RandomForest / HistGradientBoosting — same conclusion for direction. Logistic won for volatility with calibrated probabilities, interpretability, and 130/130 wins prove complexity was not needed.

Q: 14. What is walk-forward evaluation? Why not random train-test split?

A: Random split shuffles future into train = leakage, gives fake 60% accuracy. We do expanding walk-forward: train days 1–10 test day 11, train 1–11 test day 12, ... 26 folds per asset, 130 total. Plus

locked holdout from 2026-08-10 scored once. This mimics real deployment.

Q: 15. Why balanced accuracy, not just accuracy?

A: Intraday UP/DOWN is imbalanced (~52/48). Plain accuracy rewards always predicting majority. Balanced accuracy = (recall UP + recall DOWN)/2. Majority baseline is always 0.50. Our direction model got 0.48–0.49 on locked — worse than coin toss. Volatility model got 0.54–0.68 — real edge.

Q: 16. Your direction result is negative. Is your project a failure?

A: No — this is the strongest point. Most projects show overfitted 90% accuracy. We followed pre-registered criteria and **froze direction as negative**: median +0.1pp, 55% win rate, loses on locked for all 5 assets. That negative result justified the pivot to volatility, which passed. Science values honest negatives.

Q: 17. Why did direction fail but volatility succeed?

A: Direction is close to efficient-market random walk at 1h. Volatility clusters — high-range hour follows high-range hour (GARCH effect). Our trailing range features capture this clustering. Result: direction +0.1pp vs volatility +13.9pp.

Q: 18. Did news sentiment help?

A: No. Phase 0: news –0.99pp. Phase 4.4 fair rematch news-on-top: RELIANCE +0.55pp / 50% win FAIL, TCS –3.03pp / 25% win FAIL. New assets had only 5–13 articles vs 1218 for RELIANCE — provider coverage blocked, not effort. Google Trends also FAIL (–0.19pp pooled).

PART E: System & Deployment

Q: 19. Explain your architecture.

A: Upstox / Marketaux / GDELT / Reddit → Time Alignment (IST, session filter) → Feature Pipeline → Target (t+1h) → ML Layer → Prediction API (src/prediction/server.py) → Dashboard + Shadow Runner (scripts/run_shadow.py). GUI never contains prediction logic.

Q: 20. What is shadow mode?

A: We do not trade live. **run_shadow.py** refreshes completed 15m candles into **data/interim/live_market_15m.csv** during NSE hours, logs predictions, and labels outcomes only after the next hour completes. Measures prospective performance without money risk. Locked research files untouched.

Q: 21. How many tests? Why so many?

A: **256 passed**. Tests for session windows, IST conversion, holidays, dedup, FinBERT, target generation, chronological splits, leakage. Finance ML fails silently on a 1-hour timezone bug — tests prevent that.

PART F: Tricky / Cross Questions

Q: 22. Limitations?

A: News coverage sparse for 3 assets, Reddit only sample data, FinBERT local and slow, no live trading, holiday JSON needs manual update, 5 large-caps only — no small-caps.

Q: 23. Future scope?

A: 1. Prospective shadow metrics. 2. Option-strategy use of volatility signal (straddle sizing). 3. Better news coverage (Upstox news API). 4. Calibration + risk layer. No direction re-tuning without new hypothesis.

Q: 24. Why not just use ChatGPT to predict stocks?

A: LLMs have knowledge cutoffs, hallucinate, offer no temporal guarantee, are non-reproducible, and cannot give calibrated walk-forward metrics. We need a deterministic, leakage-free, auditable pipeline for SEBI-sensitive use — FinBERT + logistic gives that.

Viva Tips

1. Start with the 1-minute pitch (Q2), then show the Phase 4 table (130/130) vs Phase 3.5 table (direction FAIL). Professors remember numbers.
2. If challenged on low direction accuracy, say: *'My contribution is not claiming 90% accuracy, but proving what works and what does not under locked evaluation.'*
3. For 'different from existing products', always give the 4-point answer in Q5.
4. Keep answers under 60 seconds; offer detail only if asked.

Generated for AlphaSense project review preparation.